

SUP-MIMIC: A Multi-Task Clinical Diagnosis Benchmark for Evaluating LLMs' Robustness to Contradictory Evidence

SUP-MIMIC Contributors

Abstract

Current evaluations of large language models (LLMs) primarily focus on factual knowledge retrieval, overlooking the fundamental challenge of navigating the complex, non-bijective mappings between clinical indicators and diagnoses. Existing benchmarks fail to assess whether large language models truly possess the reasoning capability required for diagnostic ambiguity scenarios, where identical clinical presentations may correspond to different etiologies, and diagnostic convergence scenarios, where heterogeneous symptoms ultimately indicate the same disease. To address this issue, we propose the SUP-MIMIC, a multi-task framework utilizing MIMIC-IV-v3.1 that comprises Basic Assessment (BA), Diagnostic Divergence Task (DDT), and Diagnostic Convergence Task (DCT). Specifically, DDT is designed to evaluate the model's "one-to-many" disambiguation capability among phenotypically similar cases, while DCT assesses the model's ability to identify "many-to-one" diagnostic patterns across different pathophysiological pathways. Comprehensive evaluation of state-of-the-art LLMs reveals substantial performance degradation on DDT and DCT compared to baseline tasks, exposing a systemic reliance on statistical shortcuts over genuine causal reasoning. Our findings further highlight a conservative bias toward "healthy" predictions, implying non-trivial risks for missed diagnoses in realistic medical settings. This work establishes a rigorous methodology for quantifying clinical reasoning robustness and provides a roadmap for enhancing the safety of language models in clinical medicine.

1 Introduction

Large language models (LLMs) have demonstrated strong performance on medical question answering and clinical diagnosis benchmarks such as MedQA (Jin et al., 2020a), PubMedQA (Jin et al., 2019), MMLU-Medical (Hendrycks et al., 2021),

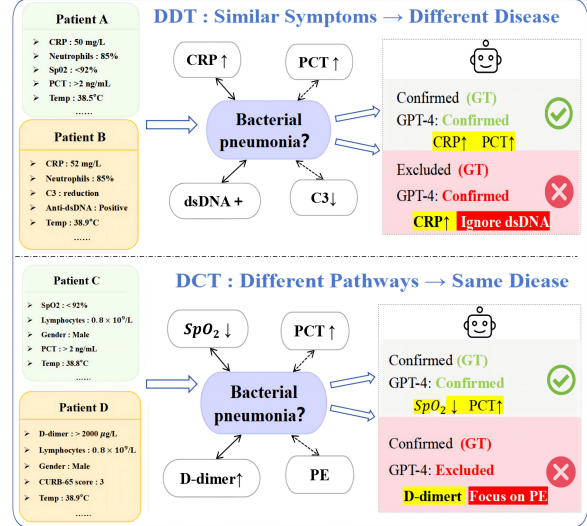


Figure 1: Illustration of two adversarial diagnostic contrasts. In Diagnostic Divergence, Patient A and Patient B share similar clinical profiles, yet the model correctly identifies bacterial pneumonia in Patient A but fails to recognize SLE in Patient B despite decisive laboratory differences. In Diagnostic Convergence, both Patient C and Patient D have bacterial pneumonia, yet the model only recognizes the classic presentation in Patient C and misses the same diagnosis in Patient D.

and LLMEval-Med (Zhang et al., 2025). However, these evaluations predominantly assess medical knowledge recall or isolated single-case diagnosis, providing limited insight into whether models remain reliable when clinical evidence is ambiguous, conflicting, or misleading. Existing benchmarks rarely provide a systematic evaluation of LLM robustness under two pervasive clinical phenomena: *diagnostic divergence*, where clinically similar patients require different diagnoses, and *diagnostic convergence*, where heterogeneous clinical presentations correspond to the same diagnosis.

In clinical practice, patient features and diagnostic labels do not follow a simple one-to-one mapping. Patients with similar demographics, laboratory values, and comorbidity profiles may re-

quire different diagnoses because a small set of clinical indicators can change the diagnostic interpretation. Conversely, patients with markedly different presentations may still receive the same diagnosis. This feature-label misalignment creates a robustness challenge for LLMs: reliable diagnosis requires sensitivity to decisive differences among similar cases and invariance to heterogeneity that does not alter the diagnosis.

Figure 1 illustrates these two failure modes with contrastive patient pairs. In the Diagnostic Divergence case, Patient A and Patient B have highly similar clinical profiles, but only Patient A has bacterial pneumonia. GPT-4 correctly identifies pneumonia in Patient A from indicators such as *PCT* and *SpO₂*, yet incorrectly assigns a similar diagnosis to Patient B despite evidence supporting an SLE flare, including *Anti-dsDNA⁺* and *C3↓*. In the Diagnostic Convergence case, both Patient C and Patient D have bacterial pneumonia, but their clinical presentations differ substantially. GPT-4 recognizes the classic presentation in Patient C while missing the same diagnosis in Patient D, whose evidence includes *D-dimer* and *CURB-65*. These examples suggest that models relying on surface similarity can fail to maintain diagnostic consistency across heterogeneous presentations.

To address this gap, we introduce **SUP-MIMIC**, a multi-task clinical diagnosis benchmark designed to evaluate LLM robustness to contradictory evidence. We construct SUP-MIMIC from MIMIC-IV-v3.1 ICU records by applying supervised feature ranking and patient-level similarity estimation to mine naturally occurring adversarial patient pairs from real clinical data. The benchmark contains three evaluation layers. **Basic Assessment (BA)** evaluates single-case diagnosis across 200 prevalent diseases. **Diagnostic Divergence Task (DDT)** tests whether models can distinguish different diagnoses among clinically similar patients. **Diagnostic Convergence Task (DCT)** tests whether models can identify the same diagnosis across clinically heterogeneous presentations.

Our experiments show that models achieve substantially higher accuracy on BA than on DDT and DCT, indicating that conventional single-case evaluation can obscure failures under contrastive clinical evidence. We further introduce pair-aware robustness metrics that assess whether model predictions preserve the diagnostic relationship within each patient pair. These metrics reveal substantial pairwise inconsistency and a pervasive healthy pre-

diction bias, where models over-predict the absence of disease in challenging cases and thereby increase the risk of missed diagnoses.

Our contributions are as follows:

- We introduce **SUP-MIMIC**, a pairwise adversarial clinical diagnosis benchmark for evaluating LLM robustness to contradictory evidence in ICU diagnosis. We formalize two complementary diagnostic robustness tasks: DDT, which evaluates sensitivity to decisive differences among clinically similar patients, and DCT, which evaluates invariance across heterogeneous presentations of the same diagnosis.
- We develop a data-driven adversarial pair mining pipeline that combines clinically informative feature ranking with patient-level similarity estimation to construct challenging diagnosis pairs from real ICU records.
- We evaluate thirteen contemporary LLMs with pair-aware robustness metrics, showing that strong single-case accuracy can coexist with substantial pairwise inconsistency and systematic failure modes such as shortcut reliance and healthy prediction bias.

2 Related work

Medical Diagnostic Benchmark Datasets. Medical AI has advanced through high-quality clinical datasets and benchmarks (Blagec et al., 2023; Gao et al., 2023). Early datasets like MIMIC-III (Harutyunyan et al., 2019) enabled clinical prediction tasks, supporting disease prediction (Zhu et al., 2021), patient prognosis (Qian et al., 2023), and personalized treatment (Wang et al., 2023b). Standardization efforts (e.g., MIMIC-Extract (Wang et al., 2020)) facilitated cross-model comparisons, while EHRShot (Wornow et al., 2023) validated few-shot transferability. Other datasets (eICU (Pollard et al., 2018), VitalDB (Lee et al., 2022)) expanded time-series prediction and risk stratification, though evaluation frameworks remained scenario-specific.

Recent diagnostic benchmarks can be roughly divided into three groups: exam-style QA, counterfactual or contrastive diagnosis benchmarks, and consistency-based clinical evaluation. Most counterfactual diagnosis benchmarks construct contrastive cases by manually editing features or text, which is useful for isolating a specific reasoning factor but departs from the natural co-occurrence

structure of ICU records. By contrast, SUP-MIMIC mines adversarial pairs directly from real clinical data, preserving the empirical interaction between symptoms, comorbidities, and laboratory findings. This makes our benchmark complementary to intervention-style counterfactual datasets rather than redundant with them.

With NLP advancements, focus shifted to clinical text comprehension. EHRNoteQA(Kweon et al., 2024) evaluated information extraction, while MedQA(Jin et al., 2020b) and LLM-MedQA(Yang et al., 2025b) assessed medical knowledge via USMLE-style questions. Retrieval-augmented generation (RAG) improved QA accuracy by 18%(Yu et al., 2024). LLMs further spurred diagnostic evaluations through benchmarks like MedMCQA(Pal et al., 2022) and ClinicalGPT(Wang et al., 2023a).

However, existing benchmarks lack causal reasoning assessment, primarily testing surface-level pattern matching rather than pathophysiological understanding. This prevents validation of models’ ability to distinguish spurious correlations from causal relationships. To address this gap, we propose a framework defining diagnostic scenarios for causal reasoning evaluation, with an automated pipeline for adversarial dataset construction.

Medical Causal Discovery and Diagnostic Consistency Research. Medical causal discovery methods comprise two categories: statistical association analysis (e.g., temporal inference in MIMIC-IV-Causal(Li et al., 2024), imaging correlations in RadCausality-12K(Haghighat et al., 2016)) and case-based validation (e.g., UK Biobank-Causal(Lehrer and Rheinstein, 2021)). Advanced approaches integrate knowledge graphs (ClinicalGPT(Wang et al., 2023a)) and semantic similarity (DDT-LLM(Yang et al., 2025a)), reducing analysis errors by 22%. BioLinkBERT enhances causal phrase labeling(Yasunaga et al., 2022).

Diagnostic consistency evaluations follow two paradigms: rule-based checks (NCCN(Gradishar and Salerno, 2016), AHA(Bozkurt et al., 2021), WHO standards) and similarity matching (Mayo Clinic(Cooper et al., 2009), DeepLesion(Yan et al., 2018)). These are constrained by predefined rules or superficial features, failing to capture deeper commonalities.

SUP-MIMIC differs from these consistency studies in two ways. First, we evaluate both directions of clinical non-bijection: similar patients may require different diagnoses (DDT), while dissimilar

patients may still share the same diagnosis (DCT). Second, instead of checking whether a model reproduces a fixed label under paraphrase or rule matching, we test whether the model can remain robust when evidence itself is contradictory. This turns diagnostic consistency into a harder, evidence-sensitive robustness problem.

However, existing methods lack a systematic framework for evaluating deep feature-diagnosis association reasoning in complex medical diagnosis, and merely perform fragmented pattern matching without distinguishing spurious correlations from genuine associative relationships. To fill this gap, we formalize the non-bijective nature of clinical reasoning, build adversarial evaluation datasets to systematically probe LLMs’ performance on contradictory diagnostic mappings, and validate their ability to grasp and leverage implicit pathophysiological related relationships.

3 Method

We formulate clinical diagnosis as binary verification over multi-label patient records. Each patient p has clinical indicators $\mathbf{X}_p$ and ICD-coded diagnoses $Y_p \subseteq \mathcal{D}$. Given candidate diagnosis d_k , the model predicts $y_{p,k} = \mathbb{I}[d_k \in Y_p]$. The full 448-dimensional feature vector is used for LLM input, while diagnosis-specific top- K_f features support pair mining and binary patient-diagnosis labels support evaluation. This section describes data preprocessing, disease and indicator selection, similarity estimation, and adversarial pair construction, as illustrated in Figure 2.

3.1 Data Filtering and Preprocessing

We use the MIMIC-IV-v3.1 dataset¹ (Johnson et al., 2016), focusing on intensive care unit (ICU) records. We restrict clinical measurements to the first 24 hours after ICU admission and use ICD-coded diagnoses associated with the ICU stay as verification labels. This window aligns model inputs with early clinical evidence and reduces reliance on measurements observed later in the stay. We exclude records with incomplete core information or clinically implausible values. Retained features with limited missingness are imputed using mean values or temporal interpolation when repeated measurements are available, and patient records are kept only when core structured variables such as vital signs, laboratory results, and

¹<https://physionet.org/content/mimiciv/3.1/>

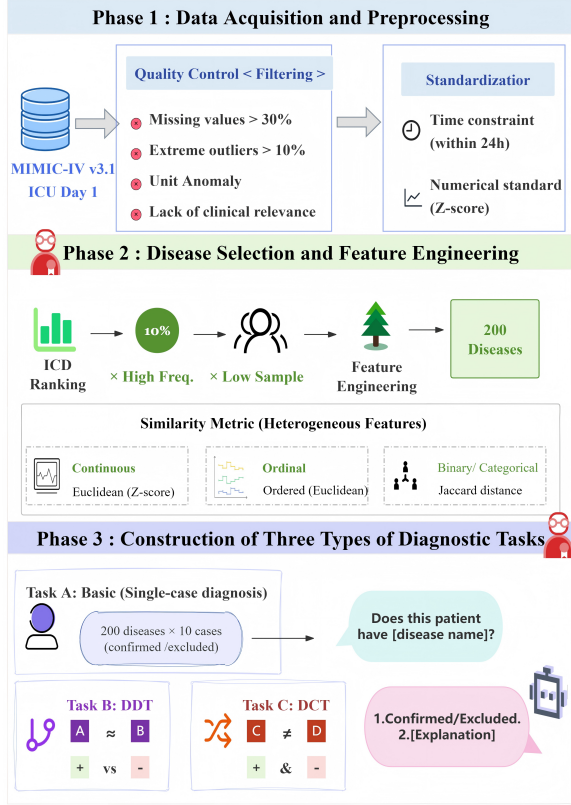


Figure 2: Overview of the SUP-MIMIC construction pipeline. Multi-label ICU records are converted into binary patient-diagnosis verification labels. The full 448-dimensional feature space is used as LLM input, while diagnosis-specific top- K_f features are used for similarity estimation and adversarial pair mining across BA, DDT, and DCT.

clinical scores are available.

3.2 Base Task Construction

We first construct the BA as the single-case diagnostic verification layer of SUP-MIMIC. We rank all ICD-coded diagnoses by frequency and exclude the top 10% most frequent labels, which tend to be broad categories that overwhelm the benchmark distribution. We also remove diagnoses with fewer than 100 positive cases and select $M = 200$ representative diseases with at least 70% data completeness and no severe outlier patterns.

For each selected diagnosis $d_k \in \mathcal{D}$, we train a Random Forest classifier to rank clinically informative features for pair mining. We train a Random Forest classifier using patients with d_k as positive examples and an equal number of randomly sampled patients without d_k as negative examples. We then rank clinical indicators by their feature importance scores and select the top K_f indicators as the diagnosis-specific feature set F_k . These feature sets are used only for patient similarity estimation and adversarial pair construction. During LLM evaluation, the model receives all clinical features rather than only the top- K_f indicators.

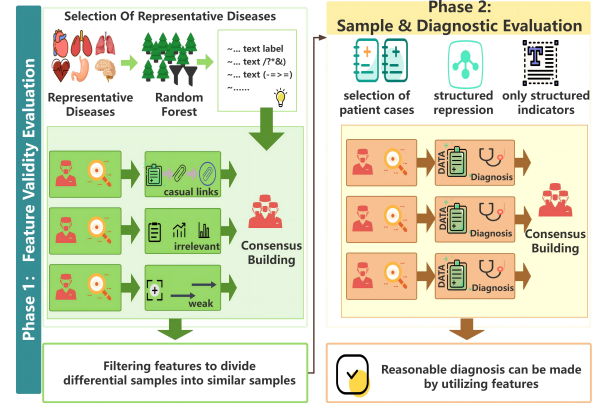


Figure 3: Diagram of the two-stage expert validation.

3.3 Case Similarity and Adversarial Pair Construction

To construct the Diagnostic Divergence Task (DDT) and Diagnostic Convergence Task (DCT), we compute patient similarity in an anchor-specific feature space. For each diagnosis d_k , similarity is measured only over its selected feature set F_k .

For patients p_i and p_j , we define

$$\text{dist}_k(p_i, p_j) = \frac{1}{|F_k|} \sum_{f \in F_k} \delta_f(x_{i,f}, x_{j,f}), \quad (1)$$

where δ_f is defined according to the feature type. For continuous variables, values are min-max normalized and compared by absolute difference:

$$\delta_f(x_{i,f}, x_{j,f}) = |z(x_{i,f}) - z(x_{j,f})|. \quad (2)$$

For ordinal variables, ordered categories are mapped to integer values, normalized, and compared in the same way. For binary or categorical variables, we use an indicator distance:

$$\delta_f(x_{i,f}, x_{j,f}) = \mathbb{I}[x_{i,f} \neq x_{j,f}]. \quad (3)$$

This per-feature formulation normalizes scale differences and prevents the distance from being dominated by any single variable type.

For each anchor diagnosis d_k , we estimate a diagnosis-specific similarity distribution by computing $\text{dist}_k(p_i, p_j)$ for patient pairs satisfying $y_{i,k} = 1$ and $y_{j,k} = 1$. Let θ_k and ϕ_k denote the 25th and 75th percentiles of this within-diagnosis distance distribution. Patient pairs are considered similar with respect to d_k if $\text{dist}_k(p_i, p_j) < \theta_k$, and highly dissimilar if $\text{dist}_k(p_i, p_j) > \phi_k$.

We then construct two sets of naturally occurring adversarial pairs. Here, ‘‘adversarial’’ refers to contrastive patient pairs mined from real clinical records where feature similarity and diagnostic labels are deliberately put in tension; it does not refer to synthetic perturbations or model-specific attacks.

For **DDT**, we mine hard negative pairs for each anchor diagnosis d_k :

$$\mathcal{P}_{\text{DDT}}^k = \{(i, j, k) : y_{i,k} = 1, y_{j,k} = 0, d_{ij}^k < \theta_k\}. \quad (4)$$

These pairs contain a patient with d_k and a clinically similar patient without d_k , testing whether models can reject the anchor diagnosis despite superficial similarity.

For **DCT**, we mine heterogeneous positive pairs for each anchor diagnosis d_k :

$$\mathcal{P}_{\text{DCT}}^k = \{(i, j, k) : y_{i,k} = 1, y_{j,k} = 1, d_{ij}^k > \phi_k\}. \quad (5)$$

These pairs contain two patients who share the anchor diagnosis but differ substantially in the anchor-specific feature space, testing whether models can recognize the same diagnosis across heterogeneous presentations.

For each diagnosis, we retain the top- m eligible pairs by distance (smallest for DDT, largest

for DCT). For a fixed anchor d_k , the two sets are disjoint by construction because they impose mutually exclusive labels on p_j . $y_{j,k} = 0$ for DDT and $y_{j,k} = 1$ for DCT. Across anchors, the same patient may appear as a distinct (p, d_k) verification instance, and all distances are computed within the corresponding F_k .

3.4 Expert Validation

We conduct two-stage expert validation to assess the clinical validity of selected features and adversarial pairs (Figure 3). In Stage 1, experts independently review disease-specific feature sets for randomly sampled benchmark diseases and categorize each feature as a diagnostic indicator, supportive but nonspecific correlate, weakly informative feature, or unclear/irrelevant feature. In Stage 2, experts review complete structured feature vectors for sampled cases and make diagnostic judgments for the target diagnosis using only the provided indicators, without free-text notes or imaging reports. This protocol assesses whether the benchmark contains clinically interpretable evidence for the intended verification tasks. Expert panel details and agreement statistics are reported in Section B.8.

4 Experiments

4.1 Evaluation Metrics

We evaluate models on three SUP-MIMIC tasks: Basic Assessment (BA), Diagnostic Divergence Task (DDT), and Diagnostic Convergence Task (DCT). Each instance is formulated as binary diagnostic verification for an anchor diagnosis d_k . Given a patient p , the model outputs $\hat{y}_{p,k} \in \{0, 1\}$, indicating whether d_k is supported by the patient’s clinical profile. For all tasks, we report overall accuracy together with Sick Recall and Healthy Recall.

Average Diagnostic Robustness (ADR). To summarize performance across standard and adversarial settings, we define ADR as the mean of task-level accuracy and pairwise robustness:

$$\text{ADR} = \frac{\text{Acc}_{\text{BA}} + \text{PDRA}_{\text{DDT}} + \text{PDRA}_{\text{DCT}}}{3}. \quad (6)$$

ADR provides a simple aggregate view of model performance across the three diagnostic verification settings.

Pairwise Diagnostic Robustness Accuracy (PDRA). Because DDT and DCT are pairwise tasks, we evaluate whether a model preserves the

Model	Acc _{BA}	Acc _{DDT}	Acc _{DCT}	ADR	PDR _{ADD}	PDR _{ADCT}	SRS	Sick _R	Healthy _R
<i>Open-Source General LLMs</i>									
Llama3.3-70B	0.3572±0.1040	0.5806±0.0701	0.3659±0.0865	0.4346±0.0452	0.2038±0.0736	0.2214±0.0659	0.1612±0.0422	0.3638±0.0441	0.7964±0.0544
Qwen2.5-7B	0.2986±0.0438	0.4995±0.0422	0.3115±0.0698	0.3699±0.0369	0.2134±0.0655	0.2934±0.1214	0.1178±0.0269	0.3025±0.0436	0.7066±0.0654
Qwen2.5-14B	0.3263±0.0375	0.5795±0.0652	0.2825±0.0601	0.3961±0.0406	0.2256±0.0655	0.3026±0.1331	0.1315±0.0244	0.3232±0.0524	0.7606±0.1227
Qwen2.5-32B	0.5619±0.1811	0.6163±0.0852	0.3462±0.0214	0.5081±0.0579	0.2694±0.1021	0.3428±0.1133	0.2567±0.0739	0.4450±0.0843	0.8240±0.0794
Mistral-7B	0.2639±0.0899	0.4927±0.0705	0.1561±0.0951	0.3042±0.0524	0.3028±0.1151	0.2802±0.1287	0.0727±0.0358	0.2266±0.0536	0.6924±0.1124
DeepSeek-V3	0.4769±0.0276	0.6070±0.0789	0.5287±0.2214	0.5375±0.0923	0.3050±0.0536	0.2956±0.1253	0.2678±0.0754	0.4796±0.0966	0.8272±0.1166
GLM-4.7	0.2564±0.1508	<u>0.6305±0.0819</u>	0.3489±0.0720	0.4119±0.0471	0.2262±0.0794	0.2530±0.1099	0.1161±0.0659	0.3318±0.0601	0.8126±0.1113
<i>Open-Source Medical LLMs</i>									
HuatuoGPT-o1-8B	0.5222±0.0719	0.6015±0.0455	0.5055±0.1201	0.5431±0.0525	0.3342±0.0829	0.3976±0.0791	0.2869±0.0629	0.4922±0.0642	0.7934±0.0548
MedReason-8B	0.4384±0.2224	0.6079±0.0486	0.4655±0.1426	0.5039±0.0819	0.2244±0.0669	0.2894±0.0555	0.2274±0.1256	0.4424±0.1094	0.8116±0.0666
<i>Closed-Source General LLMs</i>									
GPT-3.5	0.4036±0.0340	0.5290±0.0740	0.4421±0.0744	0.4582±0.0488	0.3166±0.1199	0.3450±0.1254	0.1957±0.0385	0.4036±0.0519	0.7316±0.1077
GPT-4o	0.4821±0.1265	0.6766±0.0677	0.5172±0.0037	0.5586±0.0281	0.3662±0.1488	0.2990±0.0933	0.2826±0.0641	0.4999±0.0518	0.8504±0.1071
Gemini-2.5 Flash	0.3983±0.0924	0.6184±0.0821	0.4833±0.1855	0.5000±0.0352	0.3000±0.1221	0.3236±0.1544	0.2045±0.0319	0.4420±0.0357	0.7898±0.0424
Claude Sonnet 4.5	0.2654±0.0693	0.5737±0.0867	0.4061±0.0712	0.4151±0.0533	0.3128±0.1226	0.2820±0.0911	0.1297±0.0468	0.3500±0.0594	0.7478±0.0820

Table 1: Main results on SUP-MIMIC. All metrics are reported as mean $\pm$ standard deviation over five independent runs. The best result in each column is **bolded** and the second-best is underlined.

correct diagnostic relationship within each adversarial pair. For both tasks, a pair is counted as correct only when the model makes the correct binary verification decision for both patients with respect to the same anchor diagnosis.

For DDT, each pair (p_i, p_j, d_k) contains a positive patient ($y_{i,k} = 1$) and a hard negative patient ($y_{j,k} = 0$). The pair is correct only if the model accepts d_k for p_i and rejects d_k for p_j :

$$\text{PDR}_{\text{DDT}} = \frac{1}{|\mathcal{P}_{\text{DDT}}|} \sum_{(i,j,k) \in \mathcal{P}_{\text{DDT}}} \mathbb{I}[\hat{y}_{i,k} = 1 \wedge \hat{y}_{j,k} = 0]. \quad (7)$$

For DCT, each pair contains two positive patients ($y_{i,k} = 1, y_{j,k} = 1$). The pair is correct only if the model accepts d_k for both:

$$\text{PDR}_{\text{DCT}} = \frac{1}{|\mathcal{P}_{\text{DCT}}|} \sum_{(i,j,k) \in \mathcal{P}_{\text{DCT}}} \mathbb{I}[\hat{y}_{i,k} = 1 \wedge \hat{y}_{j,k} = 1]. \quad (8)$$

SUP-MIMIC Robustness Score (SRS). To jointly evaluate standard diagnostic accuracy and adversarial pairwise robustness, we define the SUP-MIMIC Robustness Score:

$$\text{SRS} = \text{Acc}_{\text{BA}} \cdot \sqrt{\text{PDR}_{\text{DDT}} \cdot \text{PDR}_{\text{DCT}}}. \quad (9)$$

The geometric mean penalizes models that perform well on one adversarial task but poorly on the other, while Acc_{BA} ensures that the composite score remains grounded in basic diagnostic performance. We report SRS as a compact summary and analyze its components separately.

4.2 Main Results

Table 1 reports mean performance across five independent runs; the corresponding standard deviations are now provided in Appendix B and will be shown in the revised table as mean $\pm$ std. For the overall score, the standard deviations are 0.0452,

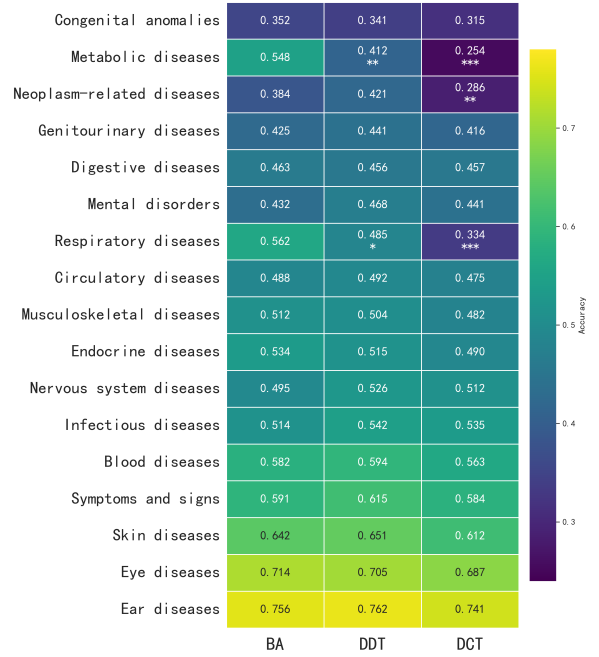


Figure 4: Accuracy heatmap for 17 clinical categories. Categories defined by distinctive single-system biomarker profiles (top rows) maintain stable accuracy, whereas multi-system categories (bottom rows) exhibit significant DCT collapse. Significance markers denote within-category paired t -tests against BA (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

0.0369, 0.0406, 0.0579, 0.0524, 0.0488, 0.0281, 0.0923, 0.0352, 0.0533, 0.0525, 0.0819, and 0.0471 for Llama3.3-70B, Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-32B, Mistral-7B, GPT-3.5, GPT-4o, DeepSeek-V3, Gemini-2.5, Claude Sonnet 4.5, HuatuoGPT-o1-8B, MedReason-8B, and GLM-4.7, respectively. Three principal findings emerge.

Pairwise evaluation exposes reasoning inconsistency masked by pointwise accuracy. Most models achieve higher accuracy on DDT than on BA because each anchor diagnosis is rare in the patient pool and models that default to disease-absent predictions benefit from this class imbalance. The

balanced construction of DDT removes this advantage at the pointwise level, yet PDRA_{DDT} drops sharply: from 0.68 to 0.37 for GPT-4o and from 0.58 to 0.20 for Llama3.3-70B. Models that appear competent on individual cases cannot reliably distinguish both members of an adversarial pair, indicating fragile rather than systematic reasoning.

Diagnostic convergence failure reflects systematic underdiagnosis. DCT accuracy remains below 0.53 for all models and falls below 0.35 for several. Because DCT requires accepting the anchor diagnosis for two clinically dissimilar patients, success depends on recognizing atypical presentations. Healthy Recall exceeds Sick Recall by 30 to 50 absolute points, confirming a pervasive bias toward rejecting the anchor diagnosis when the clinical profile deviates from prototypical patterns.

Medical pretraining improves pairwise consistency. SRS reshuffles model rankings relative to pointwise accuracy. Both medically pretrained models achieve higher SRS than general-purpose models of comparable or larger scale, and exhibit smaller Sick–Healthy Recall gaps. HuatuoGPT-o1-8B attains the highest SRS overall without leading on any individual task. Nevertheless, the best SRS of 0.29 leaves substantial room for improvement. We analyze failure patterns in subsequent sections.

4.3 Failure Mode Analysis

To understand why models fail on adversarial pairs, we classify all 128,904 incorrect predictions into four clinically grounded failure modes through structured rationale analysis.

Combinatorial Neglect (34.0%) is the dominant mode: models attend to individual clinical cues yet fail to integrate jointly decisive evidence. This error is particularly damaging in DDT, where distinguishing confusable patients requires synthesizing multiple discriminating features simultaneously. **Feature Misweighting** (23.1%) reflects miscalibrated importance assigned to clinical indicators. **Comorbidity Conflation** (22.7%) captures the inability to disentangle a primary diagnosis from coexisting chronic conditions. This mode is particularly relevant to DCT, where two patients sharing the same diagnosis present with divergent comorbidity profiles. **Biomarker Omission** (20.3%) occurs when models reference key biomarkers in their rationales but neglect them in final decisions.

The per-model error profiles trace a developmental trajectory. At the weaker end of the spectrum, models such as Mistral-7B and Qwen2.5-7B allo-

cate over 20% of errors to Biomarker Omission and exhibit elevated Comorbidity Conflation, reflecting incomplete utilization of basic clinical indicators. Stronger general models largely resolve these shortcomings but converge on a new bottleneck: Combinatorial Neglect accounts for nearly half of all errors in DeepSeek-V3 and GPT-4o, revealing that multi-evidence integration remains fragile even at frontier scale. Domain-specialized models achieve the lowest Biomarker Omission rates yet simultaneously the highest Combinatorial Neglect, confirming that biomarker recognition and evidence integration are separable competencies. Meanwhile, Feature Misweighting persists uniformly across all categories, pointing to a structural limitation that neither scaling nor domain adaptation currently addresses. Detailed patient-level error cases are omitted from the public source package pending separate data-governance review.

4.4 Degradation Analysis

Figure 5 shows that DDT and DCT degradation scores are nearly uncorrelated across models, exposing an orthogonality that pointwise accuracy alone conceals. A model that preserves performance under diagnostic divergence gains no corresponding advantage under diagnostic convergence. DeepSeek-V3 illustrates the extreme case: its DCT accuracy exceeds its own baseline while DDT degrades substantially, as though convergence and divergence reasoning draw on non-overlapping resources. HuatuoGPT-o1-8B, by contrast, maintains near-zero degradation on both tasks.

This dissociation maps cleanly onto training regime. Medical models cluster tightly at the zero-degradation origin regardless of which task is measured, achieving a task-general robustness that no individual general-purpose model replicates. General models instead distribute widely along one axis or the other, each brittle in its own idiosyncratic direction. Whether parameter scaling can substitute for this domain-acquired uniformity is the subject of Section 4.5.

4.5 Scaling Analysis

To isolate the effect of model scale from architecture and training data differences, we compare three Qwen2.5 variants that share the same design but differ in parameter count. As shown in Figure 6, scaling from 7B to 32B yields highly non-uniform returns across tasks. BA benefits most, rising from 0.30 to 0.56, an 88% relative gain that reflects im-

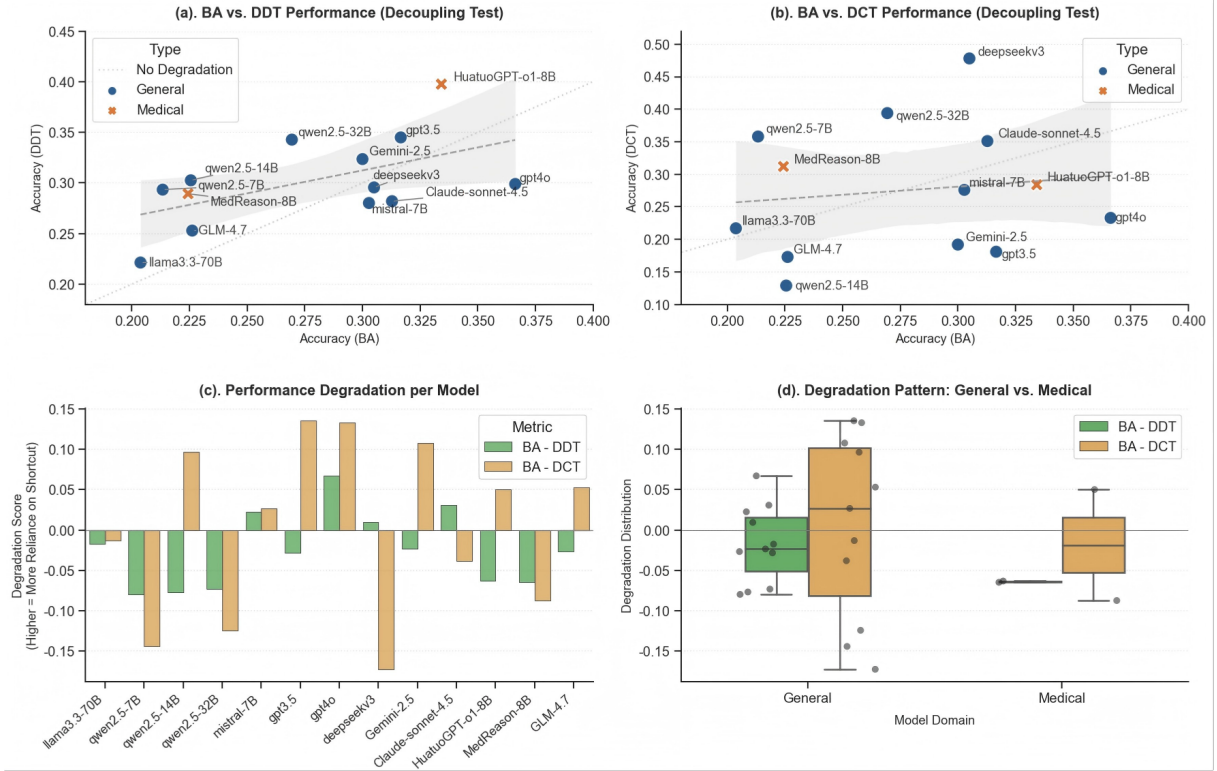


Figure 5: Performance degradation under adversarial evaluation. (a,b) Correlation between BA accuracy and task accuracy for DDT and DCT respectively. (c) Per-model degradation scores; positive values indicate performance loss. (d) Distribution of degradation scores for general-purpose versus medical models. Blue circles denote general-purpose LLMs; orange crosses denote medical LLMs; the grey dashed line marks zero degradation.

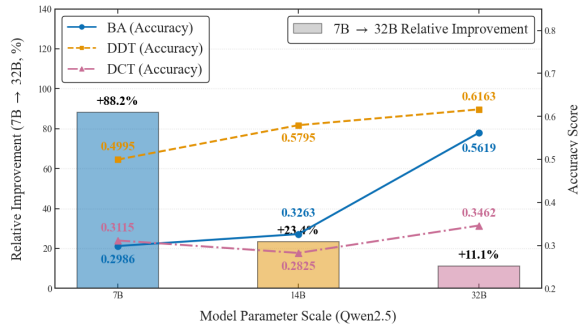


Figure 6: The Qwen2.5 series demonstrates scalability under BA, DDT, and DCT algorithms; line graphs show accuracy at 7B, 14B, and 32B resolutions; bar charts show the relative performance improvement from 7B to 32B resolution.

proved pattern recognition over the majority class. DDT improves moderately from 0.50 to 0.62, consistent with better surface-level feature matching. DCT, however, improves only marginally from 0.31 to 0.35, with a transient dip at 14B, indicating that the reasoning required to recognize atypical presentations resists parameter scaling almost entirely.

This asymmetry answers the question raised by Section 4.4: scaling amplifies the very capabilities

that BA already measures while leaving the adversarial robustness gap largely intact. The widening ratio between BA and DCT gains suggests that additional parameters preferentially consolidate prototypical pattern associations rather than cultivating the integrative reasoning that adversarial evaluation demands.

4.6 Disease Category Stratification

We map all 200 benchmark diseases to 17 ICD-based categories and test whether the degradation patterns identified above are clinically uniform. One-way ANOVA rejects uniformity for all three tasks ($F \geq 5.81$, $p < 0.001$ in each case). Figure 4 reveals a clear organizing principle. Categories defined by distinctive biomarker signatures within a single organ system, achieve stable accuracy across BA, DDT, and DCT, reflecting diagnoses that can be resolved through isolated indicator recognition. Categories requiring cross-system evidence integration, including metabolic, respiratory, and neoplasm-related diseases, show selective DCT collapse with statistical significance ($p < 0.001$ for metabolic and respiratory categories). This is precisely the setting where Combi-

natorial Neglect dominates (Section 4.3). Notably, several categories exhibit DDT accuracy exceeding BA, confirming that the class-imbalance artifact identified in Section 4.2 operates at the category level as well. Therefore, adversarial fragility is not a uniform tax on performance but a domain-selective deficit that tracks the integrative complexity of the diagnostic task.

5 Conclusion

We introduce SUP-MIMIC, a multi-task clinical diagnosis benchmark that evaluates LLM robustness to contradictory evidence through adversarial patient pairs mined from MIMIC-IV ICU records. Our experiments reveal three consistent findings: pairwise evaluation exposes reasoning inconsistency that pointwise accuracy conceals, error profiles shift from basic omission toward integration failure as model capability increases, and parameter scaling preferentially improves baseline performance while leaving adversarial robustness largely unaddressed. Medical pretraining provides more uniform robustness across tasks but does not resolve the dominant bottleneck of multi-evidence integration. These results suggest that reliable clinical diagnosis requires targeted advances in combinatorial reasoning and atypical case recognition beyond what current scaling and domain adaptation strategies provide.

Limitations

First, to ensure diagnostic reliability and minimize the impact of outliers, we excluded diagnoses with fewer than 100 ICU admissions. This is a deliberate quality-control choice rather than an omission: extremely rare diseases often exhibit heavy label noise, sparse samples, and unstable feature distributions, which would blur the benchmark signal instead of strengthening it. For example, a rare diagnosis represented by only a few dozen ICU stays may have its apparent feature profile dominated by one comorbidity cluster or by missing laboratory measurements, making adversarial pair mining unstable and difficult to clinically validate. Second, this study relied solely on structured EHR features for evaluation and did not incorporate multimodal information such as medical imaging and waveform data. That limitation is intentional for the current benchmark, which aims to isolate contradictory structured evidence at a fixed early time window. The revised manuscript also makes clear

that supplementary classical supervised analyses on the same structured features can already recover strong disease-level discrimination, while our main benchmark target is LLM robustness under contradictory evidence rather than raw classification capacity.

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A Data Access and Licensing

MIMIC-IV-v3.1 is a well-established, de-identified clinical database that has already undergone strict, professional anonymization according to HIPAA Safe Harbor standards. In this work, we utilize the MIMIC-IV-v3.1 dataset. Access to this database was approved after completion of the required safe research training (CITI program). The dataset is used in strict accordance with the PhysioNet Credentialed Health Data License. The artifacts and evaluation frameworks created in this study (SUP-MIMIC) will be released under the MIT License for non-commercial academic research purposes. During the preparation of this manuscript, AI assistants (such as ChatGPT/Claude) were utilized strictly for English language polishing, grammatical corrections, and text proofreading. The AI assistants were not involved in driving the core scientific conclusions, creating the dataset, or directly generating the primary experimental code.

B More experimental results

B.1 Feature Selection and Pair-Construction Framework

To determine how many important features are sufficient for constructing reliable evaluation pairs, we vary $K \in \{5, 10, \dots, 50\}$ and examine: (i) how many distinct original variables are covered by the union of per-disease top- K features (feature utilization), and (ii) how LLM diagnostic accuracy and stability change across diseases.

For each K , we train a random forest on the full 448-dimensional space, extract the top- K important features per disease, and take their union. The size of this union is reported as *Available Features* in Table 2, and its ratio to 448 as *Feature Utilization*. Thus, feature utilization measures how much of the original feature space is actually involved in constructing similar/dissimilar patient pairs, even though the LLM always receives all 448 features.

As shown in Table 2 and Figure 7, increasing K from 5 to 20 raises feature utilization from 1.78% (8/448) to 51.33% (230/448), and accuracy from 0.2862 to a peak of 0.3563. Small K values therefore suffer from poor coverage: pairs are repeatedly formed based on a narrow subset of clinical attributes, underutilizing the diversity of the 448-dimensional space and yielding weaker performance across the 35 diseases.

When $K > 20$, feature utilization continues to grow (57.36%–64.51%), but accuracy drops to

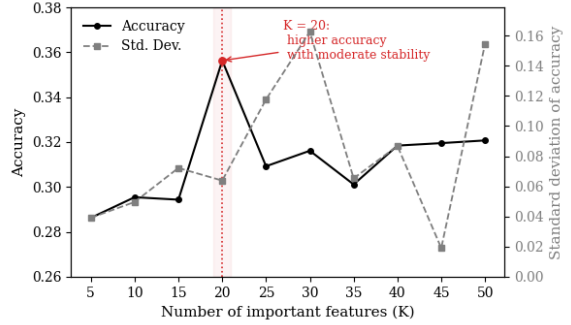


Figure 7: Trade-off between diagnostic accuracy and variance as a function of the number of important features (K), with $K = 20$ achieving the best balance.

about 0.31–0.32 and the variance across diseases increases markedly (e.g., standard deviation 0.0638 at $K = 20$ vs 0.1177 at $K = 25$ and 0.1626 at $K = 30$). This suggests that adding more “important” features beyond $K = 20$ mainly introduces marginal or noisy variables, leading to more heterogeneous and less interpretable behavior.

Overall, $K = 20$ strikes a good balance between coverage and stability. It covers over half of the original variables, achieves the highest overall accuracy, and maintains moderate cross-disease variance. We therefore adopt $K = 20$ as the default for constructing similar and dissimilar patient pairs. Importantly, this only affects pair construction: each patient is still represented by the full 448-dimensional feature vector as input to the LLM.

Important Features	Available Features	Feature Utilization	Model Accuracy	Accuracy Variance	Std. Dev. (Accuracy)
5	8	1.78%	0.2862	0.00153	0.0391
10	124	27.67%	0.2954	0.00246	0.0496
15	150	33.48%	0.2943	0.00521	0.0722
20	230	51.33%	0.3563	0.00407	0.0638
25	257	57.36%	0.3092	0.01384	0.1177
30	261	58.25%	0.3161	0.02642	0.1626
35	264	58.93%	0.3011	0.00426	0.0653
40	268	59.82%	0.3184	0.00754	0.0868
45	278	62.05%	0.3195	0.00036	0.0190
50	289	64.51%	0.3207	0.02393	0.1546

Table 2: Comparison of model performance and feature utilization across different numbers of important features. Accuracy, variance, and standard deviation are reported for the overall (ALL) setting.

B.2 Random Pairing Control

To examine whether the observed degradation is caused by a specific pair ordering or by the adversarial task definition itself, we add a label-preserving random pairing control. For DDT, each random pair contains one positive and one negative case for the same task type; for DCT, each random

pair contains two positive cases. We repeat the random pairing procedure with 100 seeds and compute pair-level accuracy in the same way as the original adversarial pairs.

Task	Original Pair Acc.	Random Pair Acc.	Difference
DDT	0.3253	0.3248	0.0005
DCT	0.1657	0.1664	-0.0007

Table 3: Label-preserving random pairing control averaged across models and runs. The maximum absolute difference across all model-task combinations is 0.0075.

The random control closely matches the original pair-level accuracy, indicating that the performance drop is not an artifact of a particular pair ordering. Instead, it reflects the difficulty of the diagnostic verification tasks under feature-label tension.

B.3 Sample Size Convergence

We further test whether the benchmark is large enough for stable model comparison. Keeping the BA/DDT/DCT ratio fixed, we draw stratified subsets of 1000, 2000, 3000, 4000, and 6000 cases. For each size, we repeat sampling 200 times and compute the mean accuracy across all model runs.

Sample Size	Mean Acc.	Bootstrap Std.	Model-run Std.
1000	0.4460	0.0017	0.0837
2000	0.4463	0.0011	0.0832
3000	0.4461	0.0008	0.0829
4000	0.4461	0.0006	0.0827
6000	0.4461	0.0000	0.0827

Table 4: Sample-size convergence under stratified re-sampling. The mean accuracy stabilizes well before the full 6000-case benchmark.

The mean estimate changes by less than 0.0003 from 1000 to 6000 cases, while bootstrap variance decreases monotonically. This suggests that SUP-MIMIC is sufficiently large for stable aggregate robustness evaluation.

B.4 Structured Supervised Baseline

As a sanity check for the structured features, we parse the available random-forest supervised reports generated during dataset construction. Across 139 disease-specific reports, the mean ROC AUC is 0.8621 with a standard deviation of 0.0779, and the median ROC AUC is 0.8500. This confirms that the 448-dimensional structured variables contain learnable diagnostic signal. However, because positives are sparse in the full ICU cohort, the default 0.5 threshold yields low positive recall. Therefore, this baseline should be interpreted as a feature-signal

check rather than a replacement for the LLM robustness benchmark.

Metric	N	Mean	Std.	Median
Accuracy	139	0.9771	0.0308	0.9900
ROC AUC	139	0.8621	0.0779	0.8500
Positive Recall	139	0.0321	0.0595	0.0100
Positive F1	139	0.0558	0.0965	0.0200
Positive Support	139	276.5971	384.0106	146.0000

Table 5: Summary of disease-specific random-forest supervised reports. Accuracy is high under class imbalance, while ROC AUC provides a more informative estimate of feature separability.

B.5 Confusion Pattern & Healthy Bias

Figure 8 illustrates the trade-off between disease sensitivity and healthy-case recognition across different LLMs. A consistent discrepancy can be observed between positive-case recall and negative-case recall, where nearly all models achieve substantially higher recall on healthy samples than on disease samples. This phenomenon indicates a systematic healthy prediction bias, suggesting that current LLMs tend to favor conservative or non-pathological predictions under diagnostic uncertainty.

Among all evaluated models, GPT-4o and HuatuoGPT-o1 achieve the highest disease recall while maintaining relatively strong healthy-case recognition, demonstrating a more balanced diagnostic capability. In contrast, smaller open-source models exhibit both lower disease sensitivity and larger performance variance, indicating weaker robustness in challenging diagnostic scenarios.

B.6 ROC/PR & Threshold Sensitivity

Finally, we analyze whether model confidence provides useful discrimination and how decision thresholds affect diagnostic risk. We construct ROC and PR curves using model confidence as the scoring function, then compute Equal Error Rate (EER) thresholds and Missed Diagnosis Rate (MDR) under both the default threshold and optimized thresholds.

AUC does not degrade from BA to DDT/DCT, with maximum $|\Delta\text{AUC}| < 0.025$. GPT-4o achieves $\text{AUC} \sim 0.83$ uniformly across tasks, indicating that confidence retains some discriminative information even when default decisions are biased. At the default threshold of 0.5, all models exhibit healthy bias. EER thresholds cluster between 0.43 and 0.48, suggesting that lower thresholds are

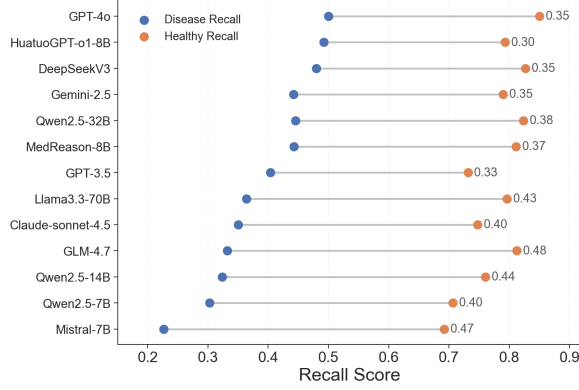


Figure 8: Diagnostic Performance Disparity Across Evaluated LLMs. The dumbbell plot in it illustrates the performance trade-offs between Disease Recall (blue circles) and Healthy Recall (orange circles). Models on the vertical axis are sorted in descending order of their Disease Recall proficiency. The horizontal grey lines represent the performance discrepancy within each model, with adjacent numerical annotations indicating the exact absolute gap.

needed to balance recall. At EER thresholds, Sick Recall improves by 10–15 percentage points, for example from 52% to 65% for GPT-4o. MDR, our clinical risk metric, is reduced by 20–25% at EER relative to the default threshold. Confidence gap analysis further shows that models assign systematically higher confidence to “healthy” predictions than to “sick” predictions.

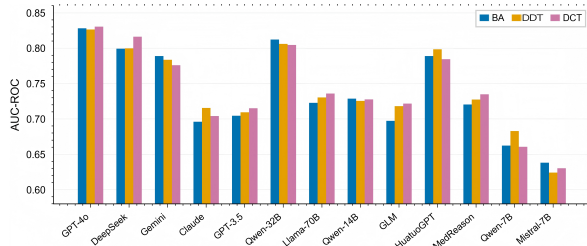


Figure 9: AUC-ROC by model and task type. Near-identical AUC across BA, DDT, and DCT confirms that confidence-based discrimination is task-independent.

B.7 Threshold-based Evidence Recalibration

Motivated by the healthy prediction bias observed above, we add a lightweight mitigation experiment. For each model run, we convert the model’s binary prediction and confidence into a disease-support score: if the model predicts “sick”, the score is its confidence; if it predicts “healthy”, the score is one minus its confidence. We then tune a decision threshold on a calibration split and evaluate the threshold on held-out adversarial cases. This does

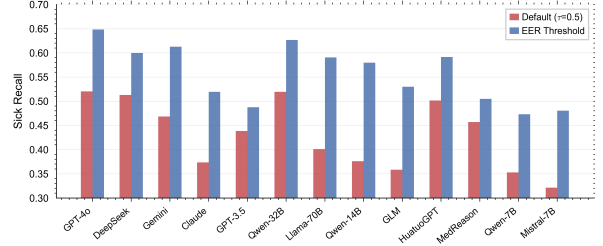


Figure 10: Sick Recall at default threshold ($\tau = 0.5$) vs EER-optimized threshold. EER calibration improves Sick Recall by 10–15 percentage points across all models.

not introduce a new model or additional clinical inputs; it tests whether explicitly recalibrating the decision boundary can reduce missed diagnoses under contradictory evidence.

Setting	Threshold	Accuracy	Balanced Acc.	Sick Recall	MDR
Default	0.500	0.4751	0.5041	0.4456	0.5544
Calibrated	0.328	0.8366	0.7436	0.9309	0.0691

Table 6: Held-out adversarial-set performance before and after threshold-based evidence recalibration. MDR denotes missed diagnosis rate.

Calibration substantially improves disease sensitivity and reduces missed diagnosis rate, but it also shifts the decision boundary toward positive predictions. Therefore, this simple pipeline is best viewed as evidence that the failure mode is partially mitigable, while robust multi-evidence integration remains an open modeling challenge.

B.8 Benchmark Quality Assessment

We convened an expert panel comprising three practicing physicians to conduct a clinical quality assessment of the benchmark dataset. The panel consisted of one associate chief physician specializing in intensive care medicine from a Grade 3A (Class A Tertiary) teaching hospital, and two attending physicians from independent hospitals. All members hold post-graduate degrees and possess at least eight years of active clinical experience. As academic co-contributors to this study, they participated voluntarily for scientific collaboration purposes, and thus external commercial compensation was not applicable.

To evaluate feature efficacy, we randomly selected five diseases from the benchmark dataset and extracted the top 20 features identified for each disease via our Random Forest feature selection pipeline. The selected diseases span major ICU-relevant categories, including infectious, cardio-

vascular, respiratory, metabolic, and oncological conditions. The three physicians independently reviewed these features without access to the ICD gold labels or pair-construction rules. The expert panel agreed that these features are tightly coupled with the core pathophysiological mechanisms of the target diagnoses. Consequently, these features were deemed highly appropriate for constructing clinically meaningful adversarial patient pairs.

We further assessed whether the 448-dimensional structured feature representation utilized in SUP-MIMIC conveys sufficient diagnostic information to sustain robust clinical reasoning. For each of the five diseases, we randomly sampled 30 patient cases, yielding 150 cases for patient-level assessment. The cases were shuffled before review, and reviewers were not informed of the ICD gold labels, pair type, or pair-construction strategy. Operating independently and without access to free-text clinical notes or imaging reports, each physician made diagnostic determinations solely using the structured clinical indicators and evaluated which specific features effectively buttressed their decisions. Because each sampled case was independently reviewed by all three physicians, every case received at least two independent clinical judgments and this stage also functions as a small-scale physician baseline for the structured verification setting. The panel unanimously agreed that reliable diagnostics could be achieved based exclusively on the 448-dimensional structured features, subsequently stratifying them into four distinct categories. To quantify inter-annotator agreement during the feature efficacy evaluation, we computed Fleiss’ κ and pairwise Cohen’s κ coefficients. The overall Fleiss’ κ reached 0.9516, demonstrating near-perfect agreement among the annotators. The pairwise Cohen’s κ coefficients were 0.9492, 0.9456, and 0.9600, respectively. These statistical results rigorously substantiate that the feature selection pipeline employed in our benchmark construction exhibits exceptional clinical reliability, consistency, and high reproducibility.

C Prompts

To illustrate the end-to-end prompting and reasoning process for a single intensive care unit (ICU) admission, we present an anonymized example of the actual request sent to the model. All patient fea-

System prompt (fixed instruction used as {system_prompt}):

You are a clinical decision support assistant specialized in critical care and infectious diseases. You will receive structured features of a patient and must decide whether the patient has a specified disease. Please strictly follow these requirements:

- 1) Base your judgment only on the provided features. Do not invent tests, history, or information that are not given.
- 2) Output exactly one complete JSON object, with no extra characters, comments, or explanations.
- 3) JSON fields and requirements:
 - **disease**: string, must equal the disease name specified by the user.
 - **prediction**: string, must be either “yes” or “no” (binary choice), even if you are uncertain.
 - **confidence**: numeric value in the range [0,1].
 - **rationale**: short string in Chinese (at most 40 characters), briefly explaining the key basis and mentioning relationships or thresholds between indicators.
 - **explanation**: longer string in Chinese, providing a detailed reasoning process. Prefer bullet-style reasoning, and cite concrete features together with their thresholds, trends, or interactions.
- 4) Consistency: prediction, rationale, and explanation must be mutually consistent. Do not use phrases like “unable to judge” or “uncertain”, and do not introduce features that are not provided.
- 5) You must output exactly one JSON object and nothing else (no comments, no prefix/suffix text).

Figure 11: System prompt used for clinical reasoning and diagnostic evaluation.

tures are directly derived from the publicly available MIMIC-IV-v3.1 critical care database. All personal identifiers have been removed, and numerical values are slightly perturbed to protect privacy while preserving clinical plausibility and relative magnitudes.

Figures 11 and 12 show representative request examples used in our experiments. Figure 11 presents the system-level instruction used to standardize the reasoning behavior of the models, while Figure 12 illustrates how structured ICU features are converted into textual inputs for inference.

This block corresponds to the actual system messages used in our inference pipeline, where {patient_text} is programmatically generated from tabular ICU features by the Python script described in the “Labels of Features” section. The generated text integrates heterogeneous clinical variables, including demographics, laboratory findings, vital signs, and so on into a unified natural-language representation for model reasoning.

D Patient-Level Error Case Availability

Detailed patient-level error cases are not included in this public arXiv source package. Although the source MIMIC records are de-identified, dense clinical feature vectors and model rationales are treated as restricted derived artifacts unless separately approved for release. The public repository instead provides the code needed to reproduce the failure-mode analysis inside an authorized local MIMIC environment.

# Example {patient_text} (field names aligned with the actual feature selection code):		
gender: male	heart_rate_min: 82	wbc_avg: 12.4
admission_age: 68.2	heart_rate_avg: 98	lactate_max: 3.2
apsiii: 61	resp_rate_max: 30	lactate_avg: 2.4
sapsii: 44	resp_rate_min: 18	crp_max: 128
sofa: 9	resp_rate_avg: 24	creatinine_max: 1.9
gcs: 13	sbp_min: 86	bun_max: 23.5
sirs: 3	sbp_avg: 102	inr_max: 1.3
lods: 6	mbp_min: 57	d_dimer_max: 1.4
charlson: 4	mbp_avg: 71	sepsis: 1
temperature_max: 38.9	spo2_min: 89	aki: 0
temperature_min: 36.2	spo2_avg: 93	diabetes: 1
temperature_avg: 37.6	wbc_max: 15.1	chronic_pulmonary_disease: 1
heart_rate_max: 126	wbc_min: 9.8	congestive_heart_failure: 0

Figure 12: Feature presentation format provided to the LLMs during inference.